

Problem Description:

You are given a rooted tree that contains N nodes. Each node contains a lowercase alphabet.

You are required to answer Q queries of type u, c , where u is an integer and c is a lowercase alphabet. The count of nodes in the subtree of the node u containing c is considered as the answer of all the queries.

Constraints

$$1 \leq N, Q \leq 10^5$$

$$1 \leq u, v \leq N$$

- c is a lowercase alphabet
- s_i is a lowercase alphabet for all $1 \leq i \leq N$
- 1 is the root node

Input format

- First line: Two space-separated integers N and Q respectively
- Second line: A string s of length N (where the i^{th} character of s represents the character stored in node i)
- Next $N - 1$ line: Two space-separated integers u and v denoting an edge between node u and node v
- Next Q lines: An integer u and a space-separated character c

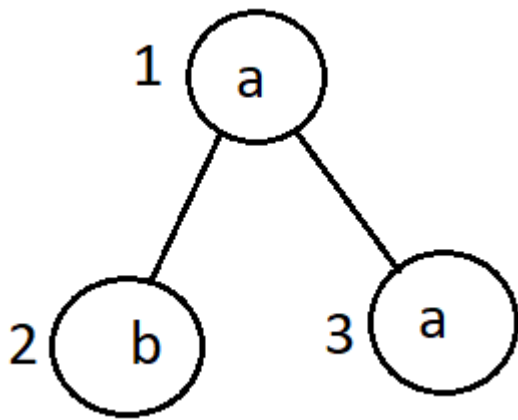
Output format

For each query, print the output in a new line.

Note: It is guaranteed that the input generates a valid tree.

Explanation for test case 1

Tree given in the sample input will look like that.



Number of nodes in the subtree of node 1 having 'a' stored in it is 2.

Logical Test Cases

Test Case 1

INPUT (STDIN)

3 1
aba
1 2
1 3
1 a

EXPECTED OUTPUT

2

Test Case 2

INPUT (STDIN)

4 1
wsx
1 2
1 3
1 1
1 w

EXPECTED OUTPUT

10

Mandatory Test Cases

Test Case 1

KEYWORD

while(Q--)

Test Case 2

KEYWORD

for(i = 0; i < N-1; i++)

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

4

Test Case 2

TOKEN COUNT

330

Test Case 3

NLOC

49